As you can see from figure 12.16, we use the priority queue to keep track of the (intermediate) reachability distances for points, and set these distances in stone only when a point is processed. Moreover, we can see that the reachability distance of the first point to be processed for each of the clusters will certainly be either undefined or larger than εMAX.

Listing 12.8 OPTICS updateQueue

```
function updateQueue(p, kdTree, queue, processed, rDists, eps,
minPts)        #1
  neighbors ← kdTree.pointsInSphere(p, eps)
                  #2
  if |neighbors| < minPoints then
                                              #3
    return queue, rDists
  for q in neighbors do
                  #4
    if q in processed then
                                                    #5
      continue
    newRDist = max(coreDistance(p, eps, minPts), distance(p,
q))              #6
    if rDists[q] == null then
                                                 #7
      rDists[q] =
newRDist                                                  #8
      queue.insert(q,
newRDist)                                                 #9
    elsif newRDist < rDist[q]
then                                              #10
      rDist[q] =
newRDist
#11
      queue.update(q, newRDist)
                                            #12
```